

Mechanics of Materials

$$\sigma_N = \frac{P}{A} \quad \sigma_B = -\frac{My}{I} \quad \tau = \frac{Tr}{I_p}$$

$$\tau_s = \frac{V}{A} \quad \tau_T = \frac{VQ}{Ib} \quad \bar{y} = \frac{\Sigma y dA}{\Sigma dA}$$

$$\varepsilon_0 = \frac{\Delta L}{L_0} \text{ (nominal)} \quad \varepsilon = \int_{L_0}^L \frac{dL}{L} = \ln(1 + \varepsilon_0) \text{ (true)}$$

$$\sigma = \frac{P}{A} \text{ (nominal)} \quad \sigma = \sigma_0(1 + \varepsilon_0) \text{ (true)}$$

$$\text{thin-wall} \quad \sigma_H(\text{hoop}) = \frac{pr}{t} \quad \sigma_L(\text{axial}) = \frac{pr}{2t}$$

For equilibrium

$$0 = \frac{\partial \tau_{xx}}{\partial x} + \frac{\partial \tau_{xy}}{\partial y} + \frac{\partial \tau_{xz}}{\partial z} + F_x$$

$$0 = \frac{\partial \tau_{yx}}{\partial x} + \frac{\partial \tau_{yy}}{\partial y} + \frac{\partial \tau_{yz}}{\partial z} + F_y$$

$$0 = \frac{\partial \tau_{zx}}{\partial x} + \frac{\partial \tau_{zy}}{\partial y} + \frac{\partial \tau_{zz}}{\partial z} + F_z$$

(Typically, $F_x = F_y = F_z = 0$)

Plane Stress

$$\sigma_{x'} = \frac{1}{2}(\sigma_x + \sigma_y) + \frac{1}{2}(\sigma_x - \sigma_y)\cos 2\theta + \tau_{xy}\sin 2\theta$$

$$\sigma_{y'} = \frac{1}{2}(\sigma_x + \sigma_y) - \frac{1}{2}(\sigma_x - \sigma_y)\cos 2\theta - \tau_{xy}\sin 2\theta$$

$$\tau_{x'y'} = -\frac{1}{2}(\sigma_x - \sigma_y)\sin 2\theta + \tau_{xy}\cos 2\theta$$

$$\sigma_{x'} + \sigma_{y'} = \sigma_x + \sigma_y \quad \tan(2\theta_p) = \frac{2\tau_{xy}}{\sigma_x - \sigma_y}$$

$$\sigma_{1,2} = \frac{\sigma_x + \sigma_y}{2} \pm \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2}$$

3D Stress

$$\det(\tau - \lambda I) = 0 \implies \lambda = \sigma_1, \sigma_2, \sigma_3$$

- Can use Mohr's circle only if 4+ shear stresses = 0
- Each circle on Mohr's circle is drawn as if looking down each principal direction axis at the element in principal stresses.

$$M = \begin{bmatrix} x' \cdot x & x' \cdot y & x' \cdot z \\ y' \cdot x & y' \cdot y & y' \cdot z \\ z' \cdot x & z' \cdot y & z' \cdot z \end{bmatrix} = \begin{bmatrix} \text{eigenvector 1} \\ \text{eigenvector 2} \\ \text{eigenvector 3} \end{bmatrix}$$

The eigenvectors describe directions of the principal stresses

$$\tau' = M\tau M^T$$

$$t = \tau n = \sigma + \tau(\text{single shear}) \quad t = \tau n = \lambda n$$

$$t - (t \cdot n)n = \tau$$

Strain

Superposition of u, v, w is valid if linearly elastic or small displacements

$$\varepsilon_x = \frac{\partial u}{\partial x} \quad \gamma_{xy} = \gamma_{yx} = \frac{\partial u}{\partial y} + \frac{\partial v}{\partial x}$$

$$\varepsilon_y = \frac{\partial v}{\partial y} \quad \gamma_{yz} = \gamma_{zy} = \frac{\partial v}{\partial z} + \frac{\partial w}{\partial y}$$

$$\varepsilon_z = \frac{\partial w}{\partial z} \quad \gamma_{xz} = \gamma_{zx} = \frac{\partial w}{\partial x} + \frac{\partial u}{\partial z}$$

$$\varepsilon = \begin{bmatrix} \varepsilon_x & \frac{1}{2}\gamma_{xy} & \frac{1}{2}\gamma_{xz} \\ \frac{1}{2}\gamma_{yx} & \varepsilon_y & \frac{1}{2}\gamma_{yz} \\ \frac{1}{2}\gamma_{zx} & \frac{1}{2}\gamma_{zy} & \varepsilon_z \end{bmatrix}$$

$$\varepsilon' = M\varepsilon M^T$$

$$e = \frac{\Delta V}{V_0} = \varepsilon_x + \varepsilon_y + \varepsilon_z = \text{dilatation}$$

Mohr's circle can be used only with tensorial strains

Strain Compatibility

Satisfy all to ensure compatibility

$$\frac{\partial^2 \varepsilon_x}{\partial y^2} + \frac{\partial^2 \varepsilon_y}{\partial x^2} = \frac{\partial^2 \gamma_{xy}}{\partial x \partial y} \quad \frac{\partial^2 \varepsilon_y}{\partial z^2} + \frac{\partial^2 \varepsilon_z}{\partial y^2} = \frac{\partial^2 \gamma_{yz}}{\partial y \partial z}$$

$$\frac{\partial^2 \varepsilon_z}{\partial x^2} + \frac{\partial^2 \varepsilon_x}{\partial z^2} = \frac{\partial^2 \gamma_{xz}}{\partial x \partial z}$$

$$2 \frac{\partial^2 \varepsilon_x}{\partial y \partial z} = \frac{\partial}{\partial x} \left(-\frac{\partial \gamma_{yz}}{\partial x} + \frac{\partial \gamma_{xz}}{\partial y} + \frac{\partial \gamma_{xy}}{\partial z} \right)$$

$$2 \frac{\partial^2 \varepsilon_y}{\partial x \partial z} = \frac{\partial}{\partial y} \left(\frac{\partial \gamma_{yz}}{\partial x} - \frac{\partial \gamma_{xz}}{\partial y} + \frac{\partial \gamma_{xy}}{\partial z} \right)$$

$$2 \frac{\partial^2 \varepsilon_z}{\partial x \partial y} = \frac{\partial}{\partial z} \left(\frac{\partial \gamma_{yz}}{\partial x} + \frac{\partial \gamma_{xz}}{\partial y} - \frac{\partial \gamma_{xy}}{\partial z} \right)$$

Hooke's Law

$$\sigma = \sigma(\varepsilon) = \tau(\varepsilon) \quad \tau = c\varepsilon \text{ (or) } \varepsilon = c^{-1}\tau$$

Uniaxial

$$\sigma = E\varepsilon \quad \nu = -\frac{\text{lateral strain}}{\text{axial strain}} \quad \varepsilon_y = \varepsilon_z = -\nu \frac{\sigma_x}{E}$$

Isotropic

$$G = \frac{E}{2(1 + \nu)}$$

$$\begin{bmatrix} \varepsilon_{xx} \\ \varepsilon_{yy} \\ \varepsilon_{zz} \\ \varepsilon_{yz} \\ \varepsilon_{zx} \\ \varepsilon_{xy} \end{bmatrix} = \frac{1}{E} \begin{bmatrix} 1 & -\nu & -\nu & 0 & 0 & 0 \\ -\nu & 1 & -\nu & 0 & 0 & 0 \\ -\nu & -\nu & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 + \nu & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 + \nu & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 + \nu \end{bmatrix} \begin{bmatrix} \tau_{xx} \\ \tau_{yy} \\ \tau_{zz} \\ \tau_{yz} \\ \tau_{zx} \\ \tau_{xy} \end{bmatrix}$$

$$\begin{bmatrix} \tau_{xx} \\ \tau_{yy} \\ \tau_{zz} \\ \tau_{yz} \\ \tau_{zx} \\ \tau_{xy} \end{bmatrix} = \begin{bmatrix} 2\mu + \lambda & \lambda & \lambda & 0 & 0 & 0 \\ \lambda & 2\mu + \lambda & \lambda & 0 & 0 & 0 \\ \lambda & \lambda & 2\mu + \lambda & 0 & 0 & 0 \\ 0 & 0 & 0 & \mu & 0 & 0 \\ 0 & 0 & 0 & 0 & \mu & 0 \\ 0 & 0 & 0 & 0 & 0 & \mu \end{bmatrix} \begin{bmatrix} \varepsilon_{xx} \\ \varepsilon_{yy} \\ \varepsilon_{zz} \\ 2\varepsilon_{yz} \\ 2\varepsilon_{zx} \\ 2\varepsilon_{xy} \end{bmatrix}$$

where $\lambda = \frac{\nu E}{(1 + \nu)(1 - 2\nu)}$ and $\mu = G$

Bulk elasticity $K = \frac{E}{3(1 - 2\nu)}$ (hydrostatic cond.)

Transverse Isotropy

Behavior is identical in directions 2 and 3, not 1

$$\frac{1}{G_{23}} = \frac{2(1 + \nu_{23})}{E_2} \quad E_2 = E_3 \quad \nu_{12} = \nu_{13} \quad G_{12} = G_{13}$$

$$\begin{bmatrix} \varepsilon_{xx} \\ \varepsilon_{yy} \\ \varepsilon_{zz} \\ \varepsilon_{yz} \\ \varepsilon_{zx} \\ \varepsilon_{xy} \end{bmatrix} = \begin{bmatrix} \frac{1}{E_1} & -\frac{\nu_{12}}{E_1} & -\frac{\nu_{13}}{E_1} & 0 & 0 & 0 \\ -\frac{\nu_{12}}{E_1} & \frac{1}{E_2} & -\frac{\nu_{23}}{E_2} & 0 & 0 & 0 \\ -\frac{\nu_{13}}{E_1} & -\frac{\nu_{23}}{E_2} & \frac{1}{E_3} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{G_{12}} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{G_{23}} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{G_{12}} \end{bmatrix} \begin{bmatrix} \tau_{xx} \\ \tau_{yy} \\ \tau_{zz} \\ \tau_{yz} \\ \tau_{zx} \\ \tau_{xy} \end{bmatrix}$$

Orthotropic

Behavior is different in each principal direction

$$\begin{bmatrix} \varepsilon_{xx} \\ \varepsilon_{yy} \\ \varepsilon_{zz} \\ \varepsilon_{yz} \\ \varepsilon_{zx} \\ \varepsilon_{xy} \end{bmatrix} = \begin{bmatrix} \frac{1}{E_1} & -\frac{\nu_{21}}{E_2} & -\frac{\nu_{31}}{E_3} & 0 & 0 & 0 \\ -\frac{\nu_{21}}{E_2} & \frac{1}{E_2} & -\frac{\nu_{32}}{E_3} & 0 & 0 & 0 \\ -\frac{\nu_{31}}{E_3} & -\frac{\nu_{32}}{E_3} & \frac{1}{E_3} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{G_{12}} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{G_{23}} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{G_{13}} \end{bmatrix} \begin{bmatrix} \tau_{xx} \\ \tau_{yy} \\ \tau_{zz} \\ \tau_{yz} \\ \tau_{zx} \\ \tau_{xy} \end{bmatrix}$$

Maxwell's recip: $\frac{\nu_{12}}{E_1} = \frac{\nu_{21}}{E_2} \quad \frac{\nu_{13}}{E_1} = \frac{\nu_{31}}{E_3} \quad \frac{\nu_{23}}{E_2} = \frac{\nu_{32}}{E_3}$

Strain Energy

$$U_0 = \frac{1}{2}(\sigma_x \varepsilon_x + \sigma_y \varepsilon_y + \sigma_z \varepsilon_z + \tau_{xy} \gamma_{xy} + \tau_{yz} \gamma_{yz} + \tau_{zx} \gamma_{zx})$$

Failure Theories

Ductile

- Max Normal: $|\sigma_1, \sigma_2, \sigma_3| \geq \sigma_y, \sigma_u$
- Max Shear Stress/Tresca: $\tau_{max} = (\sigma_1 - \sigma_3) \geq \sigma_y$
- Von Mises/Distortional Energy:

$$2\sigma_{yp}^2 = (\sigma_x - \sigma_y)^2 + (\sigma_y - \sigma_z)^2 + (\sigma_z - \sigma_x)^2 + 6(\tau_{xy}^2 + \tau_{yz}^2 + \tau_{zx}^2)$$

OR $2\sigma_{yp}^2 = (\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2$

- Octahedral Shear: $\tau_{oct} = 0.47\sigma_{yp}$
- Tresca is more conservative, VM is more accurate
- Bigger parts are more conservative

Brittle

σ_u for tensile, $\sigma_{u'}$ for compressive

- Max Normal: same as ductile, but allows $\sigma_u \neq \sigma_{u'}$
- Coulomb-Mohr: $\sigma_1/\sigma_u - \sigma_3/\sigma_{u'} = 1/n$
- Coulomb-Mohr is conservative, max normal is liberal

Table 4.3. Yield Strength σ_{yp} and Fatigue Toughness K_c for Some Materials

Metals	σ_{yp}		K_c		Minimum Values of a and t	
	MPa	(ksi)	MPa $\sqrt{m}$	(ksi $\sqrt{in.}$)	mm	(in.)
Steel						
AISI 4340	1503	(218)	59	(53.7)	3.9	(0.15)
Stainless steel						
AISI 403	690	(100)	77	(70.1)	31.1	(1.22)
Aluminum						
2024-T851	444	(64.4)	23	(20.9)	6.7	(0.26)
7075-T7351	392	(56.9)	31	(28.2)	15.6	(0.61)
Titanium						
Ti-6Al-4V	1149	(167)	66	(60.1)	8.2	(0.32)
Ti-6Al-4V	798	(116)	111	(101)	48.4	(1.91)

Cracks

$$K = \lambda \sigma_{nom} \sqrt{\pi a} \quad [MPa \cdot m^{1/2}] \quad \text{valid if } a, t \geq K_c^2 / \sigma_{yp}^2$$

Fatigue

$$N_{CR} = N_f \left(\frac{\sigma_{CR}}{\sigma_f} \right)^{1/b} \quad b = \frac{\ln(\sigma_f / \sigma_e)}{\ln(N_f / N_e)} \quad \sigma_e < \sigma_{CR} < \sigma_f$$

$$\sigma_m = \frac{1}{2}(\sigma_{max} + \sigma_{min}) \quad \sigma_a = \frac{1}{2}(\sigma_{max} - \sigma_{min})$$

Fatigue Criteria

- Modified Goodman: $\sigma_a / \sigma_{CR} + \sigma_m / \sigma_u = 1/n$
- Soderberg: $\sigma_a / \sigma_{CR} + \sigma_m / \sigma_{yp} = 1/n$
- Gerber: $\sigma_a / \sigma_{CR} + (\sigma_m / \sigma_u)^2 = 1/n$
- SAE: $\sigma_a / \sigma_{CR} + \sigma_m / \sigma_f = 1/n$

Fatigue Criterion	End points for S-N diagram (Fig. 4.3)							
	Ductile steels ($\sigma_u \leq 1750$ MPa)				Brittle (hard) steels ($\sigma_u > 1750$ MPa)			
	σ_f	N_f	σ_e	N_e	σ_f	N_f	σ_e	N_e
Modified Goodman	$0.9\sigma_u$	10^3	$\frac{1}{2}K\sigma_u$	10^6	$0.9\sigma_u$	10^3	$\frac{1}{3}K\sigma_u$	10^8
Soderberg	$0.9\sigma_u$	10^3	$\frac{1}{2}K\sigma_u$	10^6	$0.9\sigma_u$	10^3	$\frac{1}{3}K\sigma_u$	10^8
Gerber	$0.9\sigma_u$	10^3	$\frac{1}{2}K\sigma_u$	10^6	$0.9\sigma_u$	10^3	$\frac{1}{3}K\sigma_u$	10^8
SAE	$\sigma_u + 350 \times 10^6$	1	$\frac{1}{2}K\sigma_u$	10^6	σ_u	1	$\frac{1}{3}K\sigma_u$	10^8

Dynamic Loading

Vertical: $\delta_{max} = \delta_{st} \left(1 + \sqrt{1 + \frac{2h}{\delta_{st}}} \right) = \kappa \delta_{st} \quad P_{dyn} = \kappa W$

Horizontal: $\delta_{max} = \delta_{st} \sqrt{\frac{v^2}{g\delta_{st}}} \quad P_{dyn} = W \sqrt{\frac{v^2}{d\delta_{st}}}$

Theory of Elasticity

Typically, body forces equal zero.

Plane Strain

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) (\sigma_x + \sigma_y) = -\frac{1}{1 - \nu} \left(\frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y} \right)$$

Plane Stress

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) (\sigma_x + \sigma_y) = -(1 + \nu) \left(\frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y} \right)$$

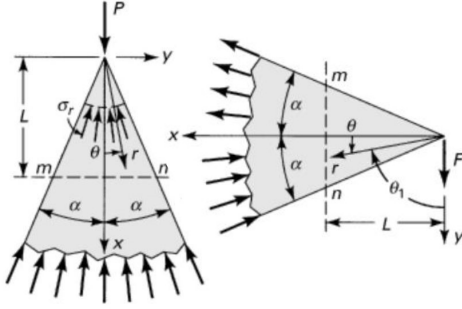
Airy's Stress Function

Define $\Phi(x, y) \longrightarrow \sigma_x = \frac{\partial^2 \Phi}{\partial y^2} \quad \sigma_y = \frac{\partial^2 \Phi}{\partial x^2} \quad \tau_{xy} = -\frac{\partial^2 \Phi}{\partial x \partial y}$

to satisfy and boundary conditions, and

$$\frac{\partial^4 \Phi}{\partial x^4} + 2 \frac{\partial^4 \Phi}{\partial x^2 \partial y^2} + \frac{\partial^4 \Phi}{\partial y^4} = \nabla^4 \Phi = 0$$

Concentrated Loads



Normal

$$\sigma_r = -\frac{P \cos(\theta)}{r(\alpha + \frac{1}{2} \sin \alpha)}$$

$$\sigma_\theta = \tau_{r\theta} = 0$$

$$\sigma_x = \sigma_r \cos^2(\theta)$$

$$= \frac{-P \cos^4(\theta)}{L(\alpha + \frac{1}{2} \sin(2\alpha))}$$

$$\sigma_y = \tau_{xy} = 0$$

Bending

$$\sigma_r = -\frac{F \cos(\theta)}{r(\alpha - \frac{1}{2} \sin \alpha)}$$

$$\sigma_\theta = \tau_{r\theta} = 0$$

$$\sigma_x = \frac{-F}{\alpha - \frac{1}{2} \sin(2\alpha)} \frac{x^2 y}{(x^2 + y^2)^2}$$

$$\sigma_y = \frac{-F}{\alpha - \frac{1}{2} \sin(2\alpha)} \frac{y^3}{(x^2 + y^2)^2}$$

$$\tau_{xy} = \frac{-F}{\alpha - \frac{1}{2} \sin(2\alpha)} \frac{xy^2}{(x^2 + y^2)^2}$$

Semi-infinite plate ($\alpha = \pi/2$):

$$\sigma_r = -\frac{2P \cos(\theta)}{\pi r} \quad \sigma_\theta = \tau_{r\theta} = 0$$

Stress Concentrations

$$K = \frac{\text{max stress}}{\text{nominal stress}} = \frac{\sigma_{max}}{\sigma_{nom}} \quad // \text{tabulated}$$

Uni-axial Tension (hole)

$$\sigma_r = \frac{1}{2} \sigma_0 \left[\left(1 - \frac{a^2}{r^2} \right) + \left(1 + \frac{3a^4}{r^4} - \frac{4a^2}{r^2} \right) \cos(2\theta) \right]$$

$$\sigma_\theta = \frac{1}{2} \sigma_0 \left[\left(1 + \frac{a^2}{r^2} \right) - \left(1 + \frac{3a^4}{r^4} \right) \cos(2\theta) \right]$$

$$\tau_{r\theta} = -\frac{1}{2} \sigma_0 \left(1 - \frac{3a^4}{r^4} + \frac{4a^2}{r^2} \right) \cos(2\theta)$$

Theory of elasticity allows for superposition if bi/tri-axial.

Elliptical hole: $K = 1 + 2(b/a) \quad \sigma_{max} = K \sigma_0 \quad 2b > 2a$

Contact Stresses

Spheres

$$P_0 = (1.5F)/(\pi a^2)$$

$$a = \left(\frac{3F}{4} \frac{(1 - \nu_1^2)/E_1 + (1 - \nu_2^2)/E_2}{1/r_1 + 1/r_2} \right)^{1/3} \quad \text{area} = \pi a^2$$

$$a = 0.88 \left(\frac{F(E_1 + E_2)r_1 r_2}{E_1 E_2 (r_1 + r_2)} \right)^{1/3} \quad \text{if } \nu_1 = \nu_2 \approx 0.3$$

$$\delta^* = 0.77 \left[F^2 \left(\frac{1}{E_1} + \frac{1}{E_2} \right)^2 \left(\frac{1}{r_1} + \frac{1}{r_2} \right) \right]^{1/3} \quad \text{if } \nu_1 = \nu_2 \approx 0.3$$

* relative displacement of sphere centers

Flat surface, $r_2 \rightarrow \infty$. Spherical cup, $r_2 \rightarrow -r_2$

Induced Stresses at Contact

Loading in z direction, z is distance from contact surface

$$\sigma_x(z) = \sigma_y(z) = -P_0 \left[\left(1 - \frac{z}{a} \operatorname{atan} \left(\frac{1}{z/a} \right) \right) (1 + \nu) - \frac{1}{2[1 + (z/a)^2]} \right]$$

$$\sigma_z(z) = \frac{-P_0}{1 + (z/a)^2} \quad \tau_{yz} = \tau_{xz} = \frac{1}{2}(\sigma_x - \sigma_z) \quad \tau_{xy} = 0$$

Parallel Cylinders

$$P_{max} = \frac{2}{\pi} \frac{F}{aL} \quad \text{area} = 2aL$$

$$a = \left[\frac{4Fr_1 r_2}{\pi L(r_1 + r_2)} \left(\frac{1 - \nu_1^2}{E_1} + \frac{1 - \nu_2^2}{E_2} \right) \right]^{1/2}$$

Induced Stresses at Contact

Loading in z direction, z is distance from contact surface

$$\sigma_x(z) = -2\nu P_0 \left(\sqrt{1 + \left(\frac{z}{a} \right)^2} - \frac{z}{a} \right) \quad \sigma_z(z) = \frac{-P_0}{\sqrt{1 + (z/a)^2}}$$

$$\sigma_y(z) = -P_0 \left[\left(2 - \frac{1}{1 + (z/a)^2} \right) \sqrt{1 + \left(\frac{z}{a} \right)^2} - 2 \frac{z}{a} \right]$$

$$\tau_{xy} = \frac{1}{2}(\sigma_x - \sigma_y) \quad \tau_{yz} = \frac{1}{2}(\sigma_y - \sigma_z) \quad \tau_{xz} = \frac{1}{2}(\sigma_x - \sigma_z)$$

General Contact

Assume/verify $\nu_1 = \nu_2$, and $E_1 = E_2$

1. Assign r_1, r'_1, r_2, r'_2 , and θ . Primes are larger radii, $0 < \theta < 180$ is between primes or non-primes

2. Calculate m, n

$$m = \frac{4}{\frac{1}{r_1} + \frac{1}{r'_1} + \frac{1}{r_2} + \frac{1}{r'_2}} \quad n = \frac{4E}{3(1 - \nu^2)}$$

3. Calculate A, B

$$A = 2/m$$

$$B = \frac{1}{2} \left[\left(\frac{1}{r_1} - \frac{1}{r'_1} \right)^2 + \left(\frac{1}{r_2} - \frac{1}{r'_2} \right)^2 + 2 \left(\frac{1}{r_1} - \frac{1}{r'_1} \right) \left(\frac{1}{r_2} - \frac{1}{r'_2} \right) \cos(2\theta) \right]^{1/2}$$

4. Calculate α , c_a , c_b (interpolate) $\cos(\alpha) = B/A$

α (deg)	c_a	c_b
20	3.778	0.408
30	2.731	0.493
35	2.397	0.530
40	2.136	0.567
45	1.926	0.604
50	1.754	0.641
55	1.611	0.678
60	1.486	0.717
65	1.378	0.759
70	1.284	0.802
75	1.202	0.846
80	1.128	0.893
85	1.061	0.944
90	1.000	1.000

5. Calculate c_a , c_b

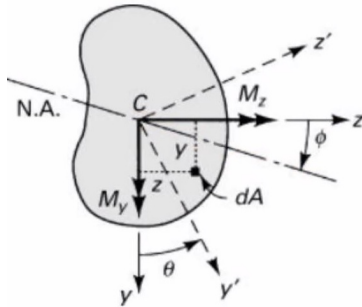
$$a = c_a \left(\frac{Fm}{n} \right)^{1/3} \quad b = c_b \left(\frac{Fm}{n} \right)^{1/3}$$

6. Calculate pressure $P_0 = (1.5F)/(\pi ab)$

Configuration	Spheres: $p_0 = 1.5 \frac{F}{\pi a^2}$	Cylinders: $p_0 = \frac{2}{\pi} \frac{F}{aL}$
A 	Sphere on a Flat Surface $a = 0.880 \sqrt[3]{Fr_1 \Delta}$ $\delta = 0.775 \sqrt[3]{F^2 \Delta^2 / r_1}$	Cylinder on a Flat Surface $a = 1.076 \sqrt{\frac{F}{L} r_1 \Delta}$ For $E_1 = E_2 = E$: $\delta = \frac{0.579F}{EL} \left(\frac{1}{3} + \ln \frac{2r_1}{a} \right)$
B 	Two Spherical Balls $a = 0.880 \sqrt[3]{F \frac{\Delta}{m}}$ $\delta = 0.775 \sqrt[3]{F^2 \Delta^2 / m}$	Two Cylindrical Rollers $a = 1.076 \sqrt{\frac{F \Delta}{Lm}}$
* note that soft object dominates results for contact area and displacement: $\Delta = \frac{1}{E_1} + \frac{1}{E_2} = \frac{E_1 + E_2}{E_1 E_2}$. If $E_1 \gg E_2$, $\Delta \approx \frac{E_1}{E_1 E_2} = \frac{1}{E_2}$		
C 	Sphere on a Spherical Seat $a = 0.880 \sqrt[3]{F \frac{\Delta}{n}}$ $\delta = 0.775 \sqrt[3]{F^2 \Delta^2 / n}$	Cylinder on a Cylindrical Seat $a = 1.076 \sqrt{\frac{F \Delta}{Ln}}$
Note: $\Delta = \frac{1}{E_1} + \frac{1}{E_2}$, $m = \frac{1}{r_1} + \frac{1}{r_2}$, $n = \frac{1}{r_1} - \frac{1}{r_2}$ * all formulae in this table assume $\nu = 0.3$		

Asymmetric Bending

Eq. in this section follow directional convention of diagram
Solve for σ_x at the point in max compression or tension



$$\sigma_x = \frac{(M_y I_y + M_z I_{yz})z - (M_y I_{yz} + M_z I_y)y}{I_y I_z - I_{yz}^2}$$

$$\tan(\phi) = \frac{y_c}{z_c} = \frac{M_y I_z + M_z I_{yz}}{M_y I_{yz} + M_z I_y} = \frac{y'_c}{z'_c} = \frac{M'_y I'_z}{M'_z I'_y}$$

Finding the right coordinate system allows for no transverse deformation, no products of inertia, only moments of inertia. Things are then symmetric, and the body behaves as expected.

$$I_y = \int_A z^2 dA \quad I_z = \int_A y^2 dA \quad I_{yz} = \int_A yz dA$$

$$\text{Where: } dA = idj \quad i, j \in x, y, z \quad i \neq j$$

Inertias Obey "Mohr's Circle"

$$I_{1,2} = \frac{I_y + I_z}{2} \pm \sqrt{\left(\frac{I_y - I_z}{2} \right)^2 + I_{yz}^2} = I_{max}, I_{min}$$

$$I'_y = \frac{I_y + I_z}{2} + \frac{I_y - I_z}{2} \cos(2\theta) - I_{yz} \sin(2\theta)$$

$$\tan(2\theta_p) = \frac{-2I_{yz}}{I_y - I_z}$$

Shear Stress

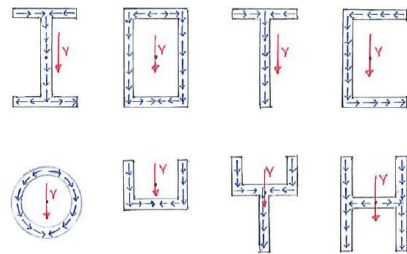
$$\tau = \frac{VQ}{It} \quad q = \frac{VQ}{I} = \tau t \quad t = \text{length of cut}$$

$$Q = (\text{Area beyond})(\text{dist from NA to centroid})$$

$$I = \sum_{i=1}^N (I_i + A_i d_i^2) \quad d_i = d_{cm} - d_{NA}$$

Shear Center

Balance internal stress and external loading to generate bending only, no torsion.



1. Resolve shear flow into stress forces along beam cross-section

$$F_{a \rightarrow b} = \int \int \tau dA \underset{\text{const. } t}{=} \int \tau t ds = \int_0^b q ds$$

2. Draw a loading with external shear force only, acting at a point on all symmetry axes, with e dimension labelled
3. Equate parts 1 and 2, solve for e

Torsion

$$\rho d\phi = \gamma dx \implies \gamma = \frac{\rho\phi}{L} \quad \phi_{\text{circ}} = \frac{TL}{GJ} \quad \tau_{\text{circ}} = \frac{T\rho}{J}$$

No warping of cross-sections, only in and out of plane:

$$u = -y\theta_z \quad v = x\theta_z \quad w = w(x, y) \quad \theta = \frac{\phi}{L}$$

Strains

$$\varepsilon_x = \gamma_{xy} = \varepsilon_y = \varepsilon_z = 0 \quad \gamma_{zx} = \frac{\partial w}{\partial x} - y\theta \quad \gamma_{zy} = \frac{\partial w}{\partial y} + x\theta$$

Stresses

$$\tau_{yz} = G \left(\frac{\partial w}{\partial y} + x\theta \right) \quad \tau_{zx} = G \left(\frac{\partial w}{\partial x} - y\theta \right)$$

$$\sigma_x = \sigma_y = \sigma_z = \tau_{xy} = 0$$

Prandtl's Stress Function

$$\text{Define } \Phi = \Phi(x, y) \longrightarrow \tau_{xz} = \frac{\partial \Phi}{\partial y} \quad \tau_{yz} = -\frac{\partial \Phi}{\partial x}$$

to satisfy and boundary conditions, and

$$\frac{\partial^2 \Phi}{\partial x^2} + \frac{\partial^2 \Phi}{\partial y^2} = H = -2G\theta$$

$$\text{Ellipse solution: } \tau_{max} = \frac{2T}{\pi ab^2} \quad \theta = \frac{(a^2 + b^2)T}{\pi a^3 b^3 G}$$

- Outer corners of shapes have zero shear stress
- Max shear stresses at free surface closest to centroid
- Circular cross-section best at holding loads

Membrane Analogy

- S is surface tension
- Stress concentrations not accounted for

Membrane Problem	Torsion Problem
z	Φ
$1/S$	G
P	2θ
$-\partial z/\partial x, \partial z/\partial y$	τ_{zy}, τ_{zx}
$2V$	T

$$\underbrace{\text{rect/unravalled open section}}_{\text{Thin-wall}} \quad \tau_{max} = \frac{3T}{bt^2} \quad \theta = \frac{3T}{bt^3 G}$$

$$\underbrace{\text{closed section}}_{\text{Thin-wall}} \quad \tau = \frac{T}{2A_m t} \quad \theta = \frac{1}{2A_m G} \oint \tau ds$$

Thick-Walled Pressure Vessels

$$\text{thick-walled if } \frac{r_i}{t} < 10$$

$$\sigma_r = \frac{a^2 P_i - b^2 P_o}{b^2 - a^2} - \frac{(P_i - P_o)a^2 b^2}{(b^2 - a^2)r^2}$$

$$\sigma_\theta = \frac{a^2 P_i - b^2 P_o}{b^2 - a^2} + \frac{(P_i - P_o)a^2 b^2}{(b^2 - a^2)r^2}$$

$$\text{if open: } \sigma_z = 0 \quad \text{if closed: } \sigma_z = \frac{P_i a^2 - P_o b^2}{b^2 - a^2}$$

$$\text{if open: } \delta = u = \frac{1 - \nu}{E} \frac{(a^2 P_i - b^2 P_o)r}{b^2 - a^2} + \frac{1 + \nu}{E} \frac{(P_i - P_o)a^2 b^2}{(b^2 - a^2)r}$$

Internal P only: $\sigma_{r,max}, \sigma_{\theta,max}$ at inner

External P only: $\sigma_{r,max}$ at outer, $\sigma_{\theta,max}$ at inner

Internal and External Pressures

$$\text{Let: } R = \frac{b}{a} \quad P = \frac{P_o}{P_i} \quad S = \frac{\sigma_{\theta_i}}{\sigma_{\theta_o}}$$

Then solve for stresses using the following ($\sigma_{\theta_o} = \sigma_\theta(r = b)$):

$$\sigma_\theta = P_i \frac{1 - PR^2}{R^2 - 1} + P_i b^2 \frac{1 - P}{(R^2 - 1)r^2} \quad S = \frac{1 + \frac{R^2(1-P)}{1-PR^2}}{1 + \frac{1-P}{1-PR^2}}$$

Press/Shrink Fits

$$\delta = \frac{bP}{E_o} \left(\frac{b^2 + c^2}{c^2 - b^2} + \nu_o \right) + \frac{bP}{E_i} \left(\frac{a^2 + b^2}{b^2 - a^2} - \nu_i \right)$$

$$P = \underbrace{\text{at surfaces}}_{\text{same material}} = \frac{E\delta}{b} \left(\frac{(b^2 - a^2)(c^2 - b^2)}{2b^2(c^2 - a^2)} \right)$$

Rotating Disks

Annular Disk

$$\sigma_r = \frac{3 + \nu}{8} \left(a^2 + b^2 - r^2 - \frac{a^2 b^2}{r^2} \right) \rho \omega^2$$

$$\sigma_\theta = \frac{3 + \nu}{8} \left(a^2 + b^2 - \frac{1 + 3\nu}{3 + \nu} r^2 + \frac{a^2 b^2}{r^2} \right) \rho \omega^2$$

$$\sigma_{r,max}(r = \sqrt{ab}) = \frac{3 + \nu}{8} (b - a)^2 \rho \omega^2$$

$$\delta = u = \frac{(3 + \nu)(1 - \nu)}{8E} \left(a^2 + b^2 - \frac{1 + \nu}{3 + \nu} r^2 + \frac{1 + \nu}{1 - \nu} \frac{a^2 b^2}{r^2} \right) \rho \omega^2 r$$

Solid Disk

$$\sigma_r = \frac{3 + \nu}{8} (b^2 - r^2) \rho \omega^2 \quad \sigma_{r,max}(r = 0) = \frac{3 + \nu}{8} b^2 \rho \omega^2$$

$$\sigma_\theta = \frac{3 + \nu}{8} \left(b^2 - \frac{1 + 3\nu}{3 + \nu} r^2 \right) \rho \omega^2 \quad \sigma_{\theta,max}(r = 0) = \frac{3 + \nu}{8} b^2 \rho \omega^2$$

$$\delta = u = \frac{1 - \nu}{8E} [(3 + \nu)b^2 - (1 + \nu)r^2] \rho \omega^2 r$$

Strain Energy

$$U_0 = \frac{1}{2}(\sigma_x \varepsilon_x + \sigma_y \varepsilon_y + \sigma_z \varepsilon_z + \tau_{xy} \gamma_{xy} + \tau_{yz} \gamma_{yz} + \tau_{xz} \gamma_{xz}) = \frac{1}{2}(\sigma : \varepsilon)$$

$$U_0 = \frac{1}{2E}(\sigma_x^2 + \sigma_y^2 + \sigma_z^2) - \frac{\nu}{E}(\sigma_x \sigma_y + \sigma_y \sigma_z + \sigma_x \sigma_z) + \frac{1}{2G}(\tau_{xy}^2 + \tau_{yz}^2 + \tau_{xz}^2)$$

$$U = \int \frac{N^2}{2AE} dx + \int \frac{M^2}{2EI} dx + \int \frac{\alpha V^2}{2AG} dx + \int \frac{T^2}{2JG} dx$$

Castigliano's Theorem

Can be applied with dummy loads ($P = 0$) at locations of interest, or applied to find reaction forces/couples in indeterminate structures.

$$\delta_i = \frac{1}{AE} \int N \frac{\partial N}{\partial P_i} dx + \frac{1}{EI} \int M \frac{\partial M}{\partial P_i} dx + \frac{1}{AG} \int \alpha V \frac{\partial V}{\partial P_i} dx + \frac{1}{JG} \int T \frac{\partial T}{\partial P_i} dx$$

$$\theta_i = \frac{1}{AE} \int N \frac{\partial N}{\partial C_i} dx + \frac{1}{EI} \int M \frac{\partial M}{\partial C_i} dx + \frac{1}{AG} \int \alpha V \frac{\partial V}{\partial C_i} dx + \frac{1}{JG} \int T \frac{\partial T}{\partial C_i} dx$$

* Applies to box beams, and channels too

α	Value
Rectangular	6/5
I-Beam*	A/A_{web}
Circle	10/9
Thin-wall cyl.	2

Buckling

$$P_{crit} = \frac{\pi^2 EI}{L_e^2} \quad r = \sqrt{\frac{I}{A}} \quad \frac{L}{r} = \text{slenderness ratio}$$

Condition	L_e
Pinned-pinned	L
Fixed-free	$2L$
Fixed-fixed	$L/2$
Pinned-fixed	$0.7L$

Plasticity

$$\sigma_0 = K \varepsilon_0^n \quad \log(\sigma_0) = \log(K) + n \log(\varepsilon) \quad \text{zero indicates true}$$

Bending Elastic-Perfectly Plastic

$$M = \frac{3}{2} M_{yp} \left(1 - \frac{1}{3} \left(\frac{e}{h} \right)^2 \right) \quad \begin{array}{l} e = \text{non-plastic region} \\ \text{rect. } b \times 2h \text{ beams only} \end{array}$$

$$M = \frac{3}{2} M_{yp} \left(1 - \frac{1}{3} \left(\frac{r}{r_{yp}} \right)^2 \right) = M_u \left(1 - \frac{1}{3} \left(\frac{r}{r_{yp}} \right)^2 \right)$$

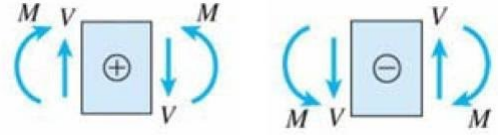
Torsion Elastic-Perfectly Plastic

$$T = \frac{\pi c^3}{6} \left(4 - \frac{\rho_0^3}{c^3} \right) \tau_{yp} = \frac{4}{3} \tau_{yp} \left(1 - \frac{1}{4} \frac{\rho_0^3}{c^3} \right) \quad \phi_{yp} = \frac{\tau_{yp} L}{Gc}$$

$$\text{at onset of yield} \quad \frac{\rho_0}{c} = \frac{\phi_{yp}}{\phi} \quad T = \frac{4}{3} \tau_{yp} \left(1 - \frac{1}{4} \frac{\phi_{yp}^3}{\phi^3} \right)$$

Other

$$\delta = \frac{PL}{EA} \quad \delta_T = \alpha \Delta T L$$



Deformation sign convention